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(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

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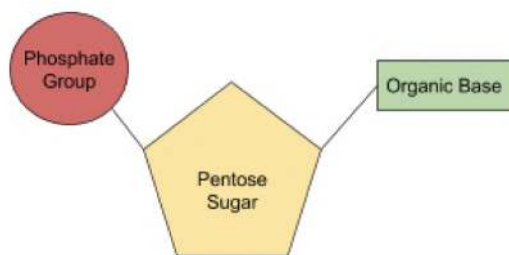
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DNA, Genes, Mutations - The Central Dogma of Biology

▼ Describe Nucleotides and the formation of DNA

- DNA and RNA are polynucleotides formed by mono-nucleotides joining together in condensation reactions to form phosphodiester bonds the phosphate group of one nucleotide and the deoxyribose or ribose sugar of another.
- Structure of mononucleotides - a deoxyribose/ribose sugar linked to a phosphate group and a base - thymine, adenine, guanine, cytosine or uracil.



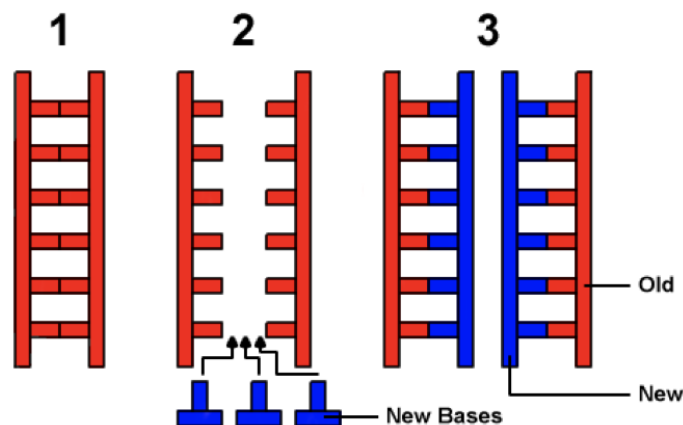
- DNA's double helix structure is formed by two polynucleotide chains running antiparallel to each other being joined together by hydrogen bonds between the two strands.
- 2 hydrogen bonds form between A & T while 3 form between C & G.
- The hydrogen bonds give DNA its stable structure as it prevents chemical bases being corrupted by outside chemicals and forces.

▼ Describe the process of DNA replication

- DNA replication is the copying of DNA to allow genetic information to be passed on from one cell/generation to another.
- DNA replication is known to be semi-conservative as each DNA molecule formed from replication contains 1 new strand and 1 of the

original parental strand. The semi-conservative nature of DNA is proven by the experiment of Meselson and Stahl.

- Steps of DNA Replication: 1. The double helix of DNA unwinds as DNA **Helicase** (enzyme) breaks down the hydrogen bonds between the complementary bases, thus separating the two parental strands. 2. One of the strands act as the template strand and complementary base pairing starts to occur as free nucleotides attach themselves to the exposed strand. 3. **DNA polymerase** then moves along the strand and lines up and catalyses the formation of phosphodiester bonds between adjacent nucleotides in a series of condensation reactions. 4. **DNA ligase** then finally catalyses the formation of phosphodiester bonds between the two DNA strands and forms a DNA double helix.



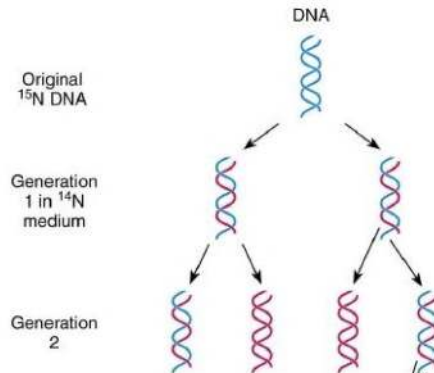
▼ Describe Meselson and Stahl's Experiment:

Source: Physics & Maths Tutor

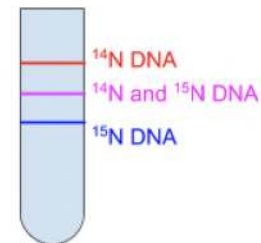
Meselson and Stahl's experiment

2 scientists named Matthew Meselson and Franklin Stahl did an experiment to prove that DNA replicated semi-conservatively. This was what they did:

1. They began by growing bacteria in a broth containing only **one isotope of nitrogen - Nitrogen-15**, a heavier isotope than the more abundant Nitrogen-14. After allowing the bacteria to grow and take the heavy nitrogen into their DNA they isolated the DNA and **centrifuged it; the DNA settles at a point equal to its density.**



2. The bacteria containing only ^{15}N DNA were transferred to a broth now containing only a ^{14}N medium. Since DNA replicates semi-conservatively, after the first replication of DNA, each double helix contained **1 ^{14}N strand, and 1 ^{15}N strand**, so settled at a different point when centrifuged. The scientists observed this was between the settling points of ^{15}N only DNA, and ^{14}N only DNA, so concluded the DNA must contain **one strand of each.**
3. They then carried out a **second replication** of the DNA, still in the ^{14}N medium, the results of the centrifuge confirmed the semi-conservative hypothesis still, with half of the DNA settling out at the ^{14}N only point, and half of the DNA settling with a combining ^{14}N and ^{15}N point.



▼ Describe the Genetic Code

The specific order of nucleotide bases on DNA make up the genetic code. Genetic code is a **triplet code** that is **non-overlapping** and **degenerate**.

- **Triplet Code:** DNA makes up the genetic code which consists of triplets of bases; each triplet of bases codes for a particular amino acid and is known as a codon. It is believed that the bases are arranged in triplets because there are only 4 bases which need to code for 20 amino acids. Three bases/triplet codes is the minimum number that gives sufficient combinations to code for the 20 proteins (gives 64 combinations $4 \times 4 \times 4$)
- **Non-overlapping:** Each triplet is only read once and triplets don't share any bases.
- **Degenerate:** More than one triplet codes for the same amino acid (code contains more info than needed). This is useful since even if a

mutation causes a change in the base sequence, the same amino acid can be made and so no effect will be had in the organism

- The genetic codes contains **start and stop codons** which either start or stop protein synthesis.

▼ Describe Protein Synthesis

Protein synthesis - transcription and translation. In translation, DNA is copied into mRNA which then leaves the nucleus and travels to a ribosome. In transcription, the copied genetic information in mRNA is used to code for amino acids that form polypeptide chains for proteins.

1. **Transcription:**

- RNA polymerase attaches to the promoter site of the DNA and begins to unzip the two DNA strands by breaking the hydrogen bonds between complementary bases. This uncoils and separates the 2 strands that now become exposed. One of the DNA strands are used as a template antisense strand to make the mRNA.
- RNA polymerase lines up the free nucleotides that attach to the template strand by complementary base pairing and form hydrogen bonds.
- Adjacent nucleotides are linked together by phosphodiester bonds in condensation reactions, thus forming a single-stranded mRNA molecule.
- Once the RNA polymerase reaches a stop codon, the mRNA detaches from the antisense strand and moves out of the nucleus through a pore and attaches to the ribosome in the cytoplasm.

2. **Translation:**

- mRNA attaches to a ribosome and tRNA collects amino acids from the cytoplasm and carries them to the ribosome. tRNA has an amino acid binding site at one end and a triplet of bases on the other that serve as the anticodon - which is complementary to the codon on the mRNA strand.
- The anti-codon and codon join by complementary base pairing of the codons. The amino acids attached to the tRNA molecules join by forming **peptide bonds through a condensation reaction.**

- The tRNA molecules **detach themselves** from the amino acids, leaving them behind as more amino acids attach to the chain. This process is repeated leading to the formation of **a polypeptide chain** until a **stop codon** is reached on mRNA, which ends the process of protein synthesis.

Question Number	Answer	Additional Guidance	Mark
3(c) *QWC	(QWC – Spelling of technical terms must be correct and the answer must be organised in a logical sequence) - reference to ribosome (attaches to mRNA) ; - idea that tRNA carries an amino acid ; - idea of {anticodon codon interaction / complementary base pairing } between tRNA and mRNA ; - formation of hydrogen bonds (between the tRNA and mRNA) ; - reference to peptide bond (between amino acids) ; (peptide bond) formed by a condensation reaction ; - idea that tRNA released from {mRNA / ribosome} ; - idea that ribosome {attaches to / detaches from / eq} {sequence / eq} on mRNA ;	QWC emphasis answer must be in a logical sequence Penalise once for point out of sequence / context IGNORE descriptions of transcription ACCEPT AA for amino acid - ACCEPT rough endoplasmic reticulum, RER - NOT amino acids unless tRNA plural - ACCEPT description of complementary base pairing - ACCEPT peptide link - ACCEPT by an enzyme - ACCEPT ribosome moves along mRNA, a start codon / AUG, stop codon / UAA / UAG / UGA	(5)

▼ Describe how Mutations work

Mutations occur from permanent changes in the sequence of bases of DNA made during replication. Proteins are coded for by the sequence of bases on DNA, the change in base sequence can affect the polypeptide produced as **different amino acid means different folding and bonding so a different tertiary structure and 3D shape**. In some cases this can result in **non-functional proteins** and lead to **cancer**, for instance if a protein that slows cell division is non-functional, this can lead to the formation of a cancerous tumour. **Genetic disorders** are also often caused by **mutation**, such as the mutation which leads to production of sticky mucus and causes **cystic fibrosis or sickle cell anaemia**, in which a mutated form of haemoglobin distorts the shape of red blood cells.

The majority of mutations are **not harmful** however, due to the nature of the genetic code, and the presence of **non-coding introns**. As introns do not code for proteins so if a mutation occurs here, no polypeptide will be affected. There are several types of mutation:

- Substitution** - this is where **one base is substituted for another**. For instance, the sequence ATGGCA to ATCGCA. In some cases, the mutation may not change the amino acid coded for, since the

degenerate nature of the genetic code means multiple codons code for the same amino acid. Sometimes however it does result in a different amino acid.

- **Insertion** - this is where **an extra base (or bases) is inserted into the DNA sequence**. For instance GGCTAC to GGCTTTAC. This mutation often results in a **non-functional** protein as **all of the codons after the insertion are affected** as the code is non-overlapping. This is known as **frame shift**.
- **Deletion** - this is where **a base (or bases) is removed from the sequence**. For instance, TGCAAC to TGAAC. This again results in **frame shift** and a likely **non-functional** protein as all codons after the deletion are changed.

▼ Key Terms

- Gene - a sequence of bases on a DNA molecule that codes for a sequence of amino acids in a polypeptide chain
- Allele – a form of a gene.
- Genotype – The genetic composition of an organism; the combination of alleles they have
- Phenotype – The physical characteristics expressed by an organism due to its genotype
- Recessive – The allele that must be present twice in the genotype to be expressed in the phenotype
- Dominant – The allele that only needs to be present once in the genotype to be expressed in the phenotype
- Codominance – Alleles in the genotype that both contribute to the phenotype. For example the genotype AB for blood group results in the phenotype AB
- Homozygote – When an organism has 2 of the same alleles for a characteristic
- Heterozygote - When an organism has 2 different alleles for a characteristic

▼ Genetic Disorders

Sex-linked Characteristics

Sex-linked genetic disorders are caused by genes present on the sex chromosomes (X/Y). Example: Red-green colour blindness is a genetic disorder/characteristic caused by a recessive allele on the X chromosome. Thus, it affects more men who only need one of the allele present on its only X chromosome to have it.

Cystic Fibrosis

- Caused by a faulty, recessive allele on chromosome 7. The faulty gene produces a non-functional CFTR channel protein which is responsible for transporting chloride ions out of cells.
- In the respiratory tract of a healthy person, chloride ions are transported out of cells through the CFTR channel protein, into the mucus lining the epithelium. In the process, the water potential of the mucus decreases, causing water to move out of the cell via osmosis. This makes the mucus sticky enough to trap pathogens that pass through the respiratory tract when we breath.
- People with cystic fibrosis do not have the movement of chloride ions out of the cell due to the non-functional channel protein, so water does not move out of the cell either, this makes the mucus too thick and sticky meaning the cilia cannot move. This results in pathogens getting stuck and remaining in the tract and often causing infections.
- Reproduction: The cervical mucus thickens as the non-functional CFTR channel proteins cannot properly transport chloride ions out of the cell and thus water does not leave the cell either. The thickened cervical mucus prevents sperm cells from getting through to the uterus, thus making pregnancy hard. Similarly, in males, the thick mucus can block the vas deferens making it hard to transport sperm to the penis.
- Digestion: Thick mucus can block the pancreatic duct, which would prevent digestive enzymes from passing into the small intestine, so less food is broken down and absorbed, preventing normal growth. In the long term, the digestive enzymes not released from the pancreas can breakdown the pancreas, if the cells producing insulin are affected, patients can also develop diabetes.

▼ Genetic Screening

Genetic screening is a branch of gene technology which involves testing people's DNA for the presence of certain genetic disorders, like cystic

fibrosis. This allows the identification of the specific gene causing a disorder and allows the most effective treatment to be chosen.

Uses of Genetic Screening:

1. The identification of carriers - If a parent is a carrier, they can be likely to pass on the disorder to their children. This helps parents make informed decisions about having children.
2. Preimplantation genetic diagnosis (PGD) - If 2 parents have alleles for genetic disorders, to prevent passing it on to the child, they can produce the embryos by in-vitro fertilisation in a lab, then screen the embryos for ones that are healthy to be implanted into the mother.
3. Prenatal testing - The testing of genetic disorders and incorrect numbers of chromosomes once the fetus is developing. Allows parents to make decisions on whether to continue the pregnancy if child has a genetic disorder. Methods for prenatal testing:
 - Amniocentesis - Amniotic fluid containing fetal tissue is collected between 14-20 weeks via a needle through the wall of the abdomen. The fetal cells are then cultured for 2-3 weeks before screening for genetic disorders.
 - Chorionic Villus Sampling - Tissue sample is taken from the placenta between 10-14 weeks using a syringe via the cervix. The tissue is then tested immediately.

Ethical Concerns of GS —

Religion - Some religions view life as sacred and thus believe that the baby should live even if they have a genetic disorder.

Moral - Genetic screening can sometimes result in false-positive or false-negative results, meaning potentially a foetus could be aborted due to an incorrect diagnosis, leading to unnecessary loss of lives.

Social - Aborting babies with disorders after genetic screening raises the idea that those with disabilities are worth less than those without.

▼ Outline the procedure of gene therapy

1. The faulty gene is identified
2. The normal allele is cut out using special enzymes
3. Many copies of the allele are made

4. The normal allele is inserted into the target cells via vectors such as virus, plasmids, liposome, etc.,
5. The vectors are injected into the target cells
6. The normal gene is translated for the functioning protein is produced

▼ Example Exam Questions:

▼ State the differences between Replication and Transcription

Question Number	Answer	
2 (d)	Any three from:	
	replication	transcription
1	uses DNA nucleotides	uses RNA nucleotides / eq ;
2	uses DNA polymerase / eq	does not use DNA polymerase / uses RNA polymerase ;
3	reference to semi-conservative	not semi-conservative / eq ;
4	(copies) both DNA strands / eq	(copies) only {one strand / template / gene / eq} ;
5	makes DNA double helix / eq	Makes single strand mRNA / eq ;

▼ Why is a triplet code needed for protein synthesis?

Question Number	Acceptable Answers	Additional Guidance	Mark
4(b)	1. only four bases ; 2. need to code for 20 / more than 16 amino acids ; 3. three bases is the minimum number that provides sufficient combinations to code for each amino acid ;	Accept idea that 2 bases would give only 16 possibilities	(3)

▼ How does Meselson and Stahl's experiment provide evidence for the semi-conservative nature of DNA replication?

Question Number	Acceptable Answers	Additional Guidance	Mark
*4(c) QWC	Take into account quality of written communication when awarding the following points. Clarity of expression. 1. (Meselson and Stahl's experiment was designed) to distinguish between different types of replication; 2. at the start of the experiment cells were grown in (medium with) { heavy nitrogen / ^{15}N } ; 3. only a heavy DNA band was observed ; 4. the cells were then transferred to (medium with) { light nitrogen / ^{14}N } ; 5. after one round of replication an intermediate DNA band was observed ; 6. after two rounds of replication intermediate and light bands were observed ; 7. showing that replication was { semiconservative / one strand of original DNA and one strand of newly synthesised DNA } ;	Accept named types e.g between semiconservative and conservative replication Accept annotated drawing of results for MP3, 5 and 6 Accept generation in place of round of replication	(5)

▼ Explain why a gene mutation can result in an enzyme that is unable to break down lipids.

Question Number	Answer	Additional Guidance	Mark
4(c) *QWC	(QWC – Spelling of technical terms must be correct and the answer must be organised in a logical sequence) 1. idea that there is a change in the {DNA sequence / base sequence of a gene / eq } ; 2. change in amino acid / change in primary structure of { protein / enzyme } ; 3. reference to different R groups ; 4. leading to different {type / position / eq} bonding ; 5. idea of change in folding e.g. different 3D structure ; 6. idea of change in {shape / properties} of the active site ; 7. idea of {lipid / substrate / eq} does not fit in the enzyme's active site ;	QWC emphasis clarity of expression 1. IGNORE mRNA 4. ACCEPT named bond e.g. hydrogen, ionic, disulphide NOT peptide 5. ACCEPT change to tertiary structure 7. ACCEPT no enzyme-substrate complex made	(5)



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